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CHAPTER 1 • NUMBER • TOPIC 11

Using a Calculator

Type it in the right way, write the full display, then check it with a quick estimate. Brackets, the fraction key and DEG mode do the rest.



$$\frac{3}{10} = 0.3 =$$

30%**4**

VARIANTS

0

PAST PAPERS

32

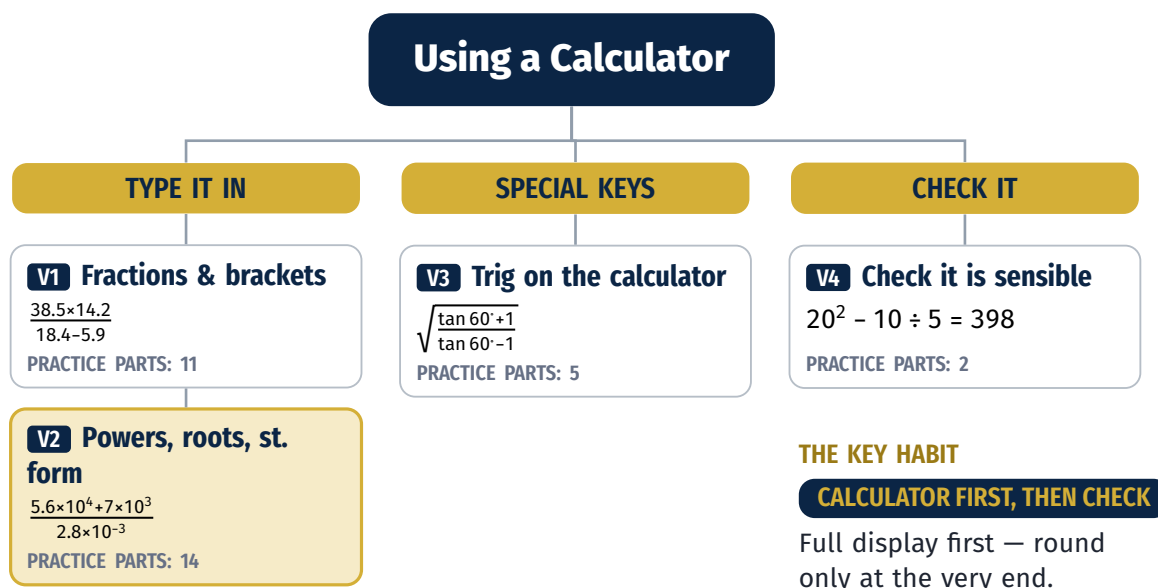
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Fractions & brackets**V2** Powers, roots & standard form**V3** Trig on the calculator**V4** Check it is sensible

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Using a calculator at a glance



Your notes


★ How often does each variant come up?

No past-paper question is only about the calculator — it is used inside almost every Paper 2 question. 32 practice parts, grouped by type.

	PRACTICE PARTS	
V1 Fractions, brackets		11
V2 Powers, roots, standard form		14
V3 Trig on the calculator		5
V4 Check it is sensible		2

WHAT THIS TELLS YOU You never get a “calculator question” — but every Paper 2 mark depends on typing it in correctly. A lost bracket turns 43.736 into 23.81 and every later mark goes with it.

BEFORE YOU START

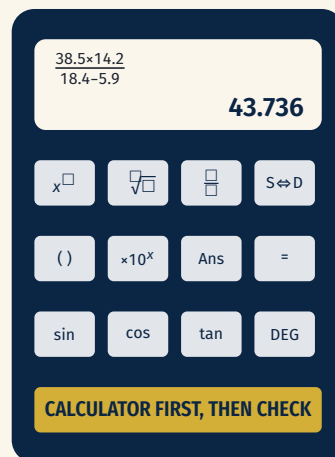
The big idea



The calculator does exactly what you type.

It does not know where the fraction line is. You tell it with **brackets** or the **fraction key**.

Then: write the **full display**, round **only at the end**, and **estimate** to check.



The keys that matter

Fraction key



Top and bottom go in their own boxes — no brackets needed.

Brackets

() ÷ ()

Use them when you type on one line.

S⇌D

$\frac{33}{4} \leftrightarrow 8.25$

Fraction or decimal: the question decides.

×10^x

2.8×10^{-3}

One key keeps the number in one piece.

Words in the question

“all the figures”

43.736

Copy the whole display.
No rounding.

“as a decimal”

$\frac{33}{4} \rightarrow 8.25$

Press S⇌D if a fraction shows.

“correct to ...”

$43.736 \rightarrow 40$

Round the full display, last.



CALCULATOR FIRST, THEN CHECK

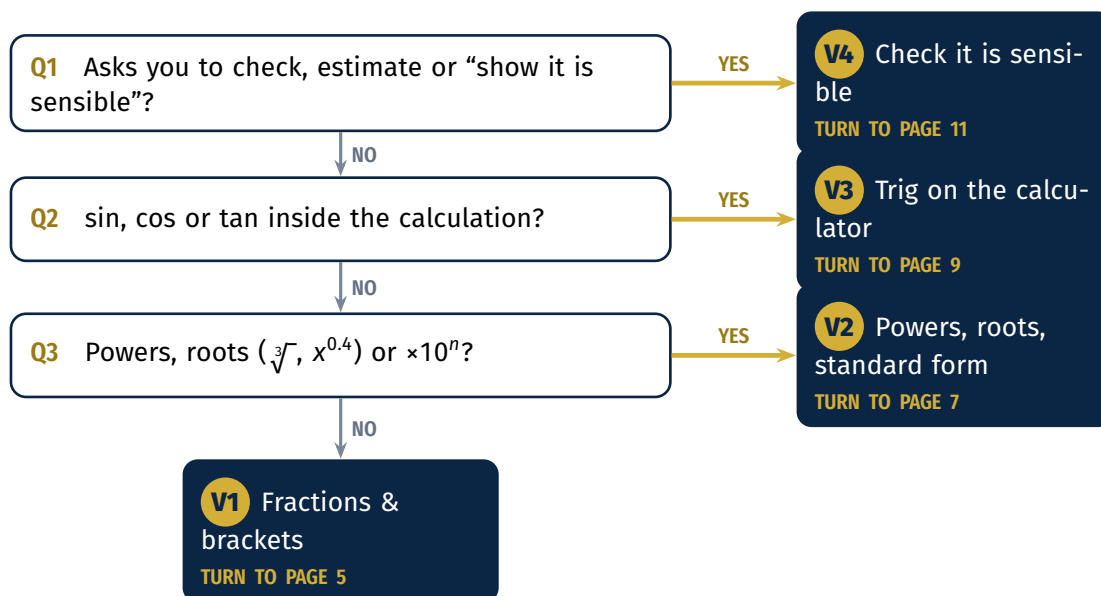
Calculator first, then check: the calculator finds the answer, a 1 s.f. estimate tells you if you typed it in right.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|--|-----------------|--|-----------------------|
| | What is it? | | When you see → you do |
| | How do I do it? | | Watch out |
| | Worked example | | Exam tip |
| | Check yourself | | Calculator check |

V1**VARIANT 1**

Fractions & Brackets

Tell the calculator where the fraction line is: fraction key, or brackets round top and bottom.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 11


Your notes

? What is it?

In plain words

A long fraction line means “work out **all of the top** ÷ **all of the bottom**”.

Type it with the **fraction key**, or put **brackets** round the top and round the bottom. Then write **all the figures** before rounding.

IT LOOKS LIKE

$$\frac{38.5 \times 14.2}{18.4 - 5.9}$$

$$\frac{\sqrt{20.4}}{6.2 \times 0.48}$$

$$\frac{38.5 \times 14.2}{18.4 - 5.9}$$

↓ **BRACKET TOP AND BOTTOM**

$$(38.5 \times 14.2) \div (18.4 - 5.9)$$

$$= 43.736 \checkmark$$

How do I do it?

1 Structure

Spot the fraction line: what is on top, what is on the bottom?

2 Type

Fraction key — or (top) ÷ (bottom).

3 Full display

Write every figure: 43.736.

4 Round last

Only now round as the question asks.

BRACKET THE TOP • BRACKET THE BOTTOM Or use the fraction key.

V1 Fractions & Brackets

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Write down all the figures on your calculator display”

→ **Copy the whole display — no rounding**

“You must give your answer as a decimal”

→ **S ⇌ D if a fraction shows**

“(b) Write your answer to part (a) correct to ...”

→ **Round the part (a) display**

WORKED EXAMPLE **PRACTICE**

Practice • Easy • Q1 • 3 marks

(a) Work out $\frac{38.5 \times 14.2}{18.4 - 5.9}$. Write down all the figures on your calculator display. (b) Write your answer to 1 significant figure.

STEP 1 top 38.5×14.2 , bottom $18.4 - 5.9$ ◀ structure

STEP 2 $(38.5 \times 14.2) \div (18.4 - 5.9)$ ◀ type with brackets

STEP 3 $= 546.7 \div 12.5 = 43.736$ ◀ full display

STEP 4 $43.736 \approx 40$ ◀ (b) 1 s.f.

ANSWER (a) 43.736 (b) 40

✓ estimate $\frac{40 \times 10}{10} = 40$ ✓

WATCH OUT

✗ $38.5 \times 14.2 \div 18.4 - 5.9 = 23.81 \dots$

✓ $(38.5 \times 14.2) \div (18.4 - 5.9)$

Without brackets the calculator only divides by 18.4.

EXAM TIP

Write the full display in (a) even if (b) asks you to round — (a) is a mark of its own.

Check yourself

calculator first • answers on page 13

1.1 Practice • Easy

Work out $\frac{\sqrt{20.4}}{6.2 \times 0.48}$.
Write all the figures on your display.

Answer:

1.2 Practice • Medium

Work out $\frac{1.45^2}{3.89 - \sqrt{5.75}}$.
Write all the figures.

Answer:

1.3 Dr Eslam

Work out $\frac{12.6 + 3.9}{2.5 \times 0.8}$ as a decimal.

Answer:



VARIANT 2

Power, Roots & Standard Form

Use the power, root and $\times 10^x$ keys; keep each number in one piece.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 14



Your notes

? What is it?

In plain words

Powers (x^2 , $x^{0.4}$), roots ($\sqrt{\quad}$, $\sqrt[3]{\quad}$, $\sqrt[5]{\quad}$) and **standard form** each have their own key.

Type 2.8×10^{-3} with the **$\times 10^x$** key so it stays **one number**. Give standard-form answers as $a \times 10^n$, $1 \leq a < 10$.

IT LOOKS LIKE

$$\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$$

$$\sqrt[3]{2^3 + 2}$$

$10^{\frac{3}{4}}$

$$\frac{63\,000}{2.8 \times 10^{-3}}$$

= 22 500 000

$$= 2.25 \times 10^7$$

$\times 10^x$ KEY: ONE NUMBER

How do I do it?

1 Keys

$x^{\square}$ for powers, $\sqrt{\square}$ for roots, $\times 10^x$ for standard form.

2 Group

Bracket the top and the bottom (or fraction key).

3 Display

Write the full answer.

4 Form

Convert to standard form or round as asked.

×10^x KEEPS IT ONE NUMBER Don't type × 10 ^.

V2 Powers, Roots & Standard Form

VARIANT 2 • SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"Give your answer in standard form" → $a \times 10^n$, $1 \leq a < 10$

$\sqrt[3]{}, \sqrt[5]{}, 10^{\frac{3}{4}}$ → **Root key with the index; fraction power in brackets**

"correct to 3 significant figures" → **Full display first, round last**

**WORKED EXAMPLE****PRACTICE**

Practice • Medium • Q5 • 2 marks

Work out $\frac{5.6 \times 10^4 + 7 \times 10^3}{2.8 \times 10^{-3}}$. Give your answer in standard form.

STEP 1 top: $5.6 \times 10^4 + 7 \times 10^3 = 63\,000$ ◀ $\times 10^x$ key

STEP 2 $63\,000 \div (2.8 \times 10^{-3})$ ◀ bottom stays one number

STEP 3 $= 22\,500\,000$ ◀ display

STEP 4 $= 2.25 \times 10^7$ ◀ standard form

ANSWER 2.25×10^7 ✓ $2.25 \times 10^7 \times 2.8 \times 10^{-3} = 63\,000$ ✓**WATCH OUT**

✗ $63\,000 \div 2.8 \times 10^{-3} = 22.5$

✓ $63\,000 \div (2.8 \times 10^{-3})$

Typed as $\times 10^{-3}$ it multiplies after dividing.

**EXAM TIP**

Count the digits: 22 500 000 has 8 digits, so the power is 10^7 .

**Check yourself**

calculator first • answers on page 13

2.1

Practice • Easy

Work out
 $4.2 \times 10^3 + 5.3 \times 10^2$ in
 standard form.

Answer:

2.2

Practice • Medium

Calculate
 $\sqrt{256^{0.25} + 4 \times 8}$.

Answer:

2.3

Practice • Hard

Find the value of
 $\sqrt[5]{3.2 \times 10^{11}}$.

Answer:

V3**VARIANT 3**

Trig on the Calculator

DEG mode, close every bracket, type the whole thing in one go.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 5


Your notes

? What is it?

In plain words

sin, cos and tan are keys. Check the screen shows **D** (degrees) — in R mode sin 30 gives -0.988.

The calculator opens a bracket after sin: **close it** before the next operation. Type the whole expression in one go — no rounding in the middle.

$$\boxed{\text{D} \quad \sin(30) \quad 0.5}$$

DEGREES MODE ✓

R MODE: sin 30 = -0.988 ×

IT LOOKS LIKE

$$\sqrt{\frac{\sin 25^\circ + \sin 40^\circ}{\cos 25^\circ - \cos 40^\circ}}$$

$$\sqrt{1 - (\sin 33^\circ)^2}$$

How do I do it?

1 Mode

Check for D on the screen.

2 Brackets

Close the bracket after each angle.

3 One go

Fraction key / root key round the whole thing.

4 Round

Full display, then round as asked.

D ON THE SCREEN Check before every trig question.

V3 Trig on the Calculator

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO $\sin$, $\cos$, $\tan$ of an angle in degrees→ **Mode D; close the bracket after the angle** $(\sin 33^\circ)^2$ → **Bracket the sin, then square**

"correct to 2 decimal places"

→ **Full display first****WORKED EXAMPLE** **PRACTICE**

Practice • Medium • Q4 • 3 marks

(a) Work out $\sqrt{\frac{\sin 25^\circ + \sin 40^\circ}{\cos 25^\circ - \cos 40^\circ}}$. Write all the figures. (b) Give your answer to 2 decimal places.

STEP 1 mode D

◀ check the screen

STEP 2 $\sqrt{\frac{\sin(25) + \sin(40)}{\cos(25) - \cos(40)}}$

◀ root key, fraction key, close brackets

STEP 3 = 2.75603957

◀ full display

STEP 4 ≈ 2.76

◀ (b) 2 d.p.

ANSWER (a) 2.75603957 (b) 2.76 ✓ top ≈ 1.07 , bottom ≈ 0.14 : $\sqrt{7.6} \approx 2.76$ ✓**WATCH OUT**✗ $\sin 30 = -0.988$ ✓ $\sin 30^\circ = 0.5$

The calculator was in radians (R). Switch to D.

EXAM TIPQuick mode test: $\sin 30$ must give 0.5.**Check yourself**

calculator first • answers on page 13

3.1 Practice • MediumWork out $\sqrt{\frac{\tan 60^\circ + 1}{\tan 60^\circ - 1}}$.
Write all the figures.

Answer:

3.2 Practice • HardWork out $\sqrt{\frac{2.5 \times \sin 43^\circ}{8.2^2 - 50.5}}$
to 3 s.f.

Answer:

3.3 Practice • HardWork out $\sqrt{1 - (\sin 33^\circ)^2}$.

Answer:

V4**VARIANT 4**

Check It Is Sensible

Round every number to 1 s.f., work it out in your head, compare with the display.

PRACTICE TOPIC
NO PAST-PAPER QUESTION OF ITS OWN
USED INSIDE OTHER QUESTIONS
PRACTICE QUESTIONS: 2

 Your notes

? What is it?

In plain words

After the calculator, **estimate**: round each number to **1 significant figure** (or a nice number like $\sqrt[3]{1000} = 10$) and work it out by hand.

If the display and the estimate are **close**, the answer is sensible. If not, you typed it wrong.

IT LOOKS LIKE

$$19.42^2 - \sqrt[3]{1006} \div 4.95$$

$$\frac{41.2 \times 9.8}{0.49}$$

display: 375.11 ...

estimate: $20^2 - 10 \div 5 = 398$

close ✓ sensible

CALCULATOR FIRST, THEN CHECK

How do I do it?

1 Calculator

Get the full display.

2 Round

Each number to 1 s.f. (or a nice root).

3 By hand

Work out the estimate — show it.

4 Compare

Close? Sensible. Far? Type it again.

1 S.F. EACH, THEN COMPARE Write the rounded numbers down.

V4 Check It Is Sensible

VARIANT 4 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Use approximations to check that your answer is sensible”

→ **1 s.f. each number, show the working**

“Show that this is within 0.1 of 3.14”

→ **Full display, then the gap: ≤ 0.1 ?**

$$\sqrt[3]{1006}, \sqrt{48}$$

→ **Nearest nice root: 10, 7**

WORKED EXAMPLE **PRACTICE**

Practice • Hard • Q4 • 3 marks

(a) Work out $19.42^2 - \sqrt[3]{1006} \div 4.95$. Write your full display. (b) Use approximations to check that your answer is sensible.

STEP 1 375.1121656

◀ (a) BIDMAS: power and root, then $\div$, then $-$

STEP 2 $19.42 \approx 20$, $\sqrt[3]{1006} \approx 10$, $4.95 \approx 5$

◀ round

STEP 3 $20^2 - 10 \div 5 = 400 - 2 = 398$

◀ by hand

STEP 4 375 and 398 are close

◀ compare

ANSWER (a) 375.1121656 (b) 398: sensible

✓ same size ✓

WATCH OUT

$$\times (19.42^2 - \sqrt[3]{1006}) \div 4.95 = 74.2$$

✓ only $\sqrt[3]{1006}$ is divided

BIDMAS: divide before you subtract.

EXAM TIP

“You must show your working”: write the rounded numbers, not just “ ≈ 400 ”.

Check yourself

calculator first • answers on page 13

4.1 Practice • Medium

Show that

$$4\left(1 - \frac{22}{57} + \frac{22}{85} - \frac{22}{105} + \frac{22}{117} - \frac{22}{242}\right)$$

is within 0.1 of 3.14.

Answer:

4.2 Dr Eslam

Estimate $\frac{41.2 \times 9.8}{0.49}$.

Answer:

4.3 Dr Eslam

The display shows

$$41.08 \text{ for } \frac{6.12 \times 19.8}{2.95}$$

Check it.

Answer:

CHECK YOURSELF

Answers



Every answer below was checked in Python (check_answers.py).

1.1 1.5176868

1.2 1.409102748

1.3 8.25

2.1 4.73×10^3

2.2 6

2.3 200

3.1 1.931851653

3.2 0.319

3.3 0.8386705679

4.1 3.0418 — within 0.1

4.2 $\frac{40 \times 10}{0.5} = 800$

4.3 $\frac{6 \times 20}{3} = 40$

THE WHOLE TOPIC ON ONE PAGE

Calculator cheat sheet



Your notes
V1 Fractions & brackets

(top) ÷ (bottom)

Or fraction key. Full display, round last.

V2 Powers, roots, st. form $63\,000 \div (2.8 \times 10^{-3})$ $\times 10^x$ key; $a \times 10^n$.**V3 Trig** $\sin 30^\circ = 0.5$

D mode; close each bracket.

V4 Check it $20^2 - 10 \div 5 = 398$

1 s.f. each, compare.

**TOP 3 LOST MARKS**

1. no brackets round the bottom ×
2. rounding in the middle ×
3. radians mode ×

BOTTOM LINE Bracket top and bottom • full display first • D mode • calculator first, then check.